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strawberry fields HIGH SCHOOL CHANDIGARH

SFHS RoboSummit 2026

Participant Rulebook: Rules and Regulations

For participating schools and students

30 October 2026

Strawberry Fields High School, Sector 26, Chandigarh

Six events for Grades 3 to 12. This booklet explains who can enter, what equipment to bring, how each event runs and how winners are decided.

Key information

Reporting: 8:15 a.m.

Registration closes: 27 October 2026

Participation fee: Rs. 500 per participant

How to use this booklet

Read the general rules first, then the section for your event: equipment limits come first, followed by competition procedure and scoring. Registration links and the final packing check are in Section 9.

Technical planning and coordination support: INNVERSE

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1. Your day at RoboSummit

Events, categories and schedule

Event	Eligibility	Team	Venue
Balloon Car Race	Grades 3-5	1-3 students	Natyashala
Hydraulic Arm	Grades 6-8	1-3 students	Einstein Hall
Line Follower	Middle and Senior	1-3 students	Newton Hall
Robo Race	Middle and Senior	1-3 students	Newton Hall
Robo Soccer	Middle and Senior	3 students / 3 robots	Globe Theatre
Drone Quidditch	Middle and Senior	1-3 students	Basketball Court

Middle: Grades 6-8. Senior: Grades 9-12. Middle and Senior results are ranked separately. All members of a team must be eligible for its registered category.

Competition day schedule

Time	Programme
8:15 a.m.	Arrival and Reporting
9:00 a.m.	Inauguration - Einstein Hall
9:30 a.m.	Events Begin
2:00 p.m.	Prize Distribution and Closing Ceremony
2:30 p.m.	Dispersal

- Each student may enter one event only.
- Schools may enter any number of teams and students across eligible events.
- Refreshments will be provided to students and accompanying educators.

2. General rules

These rules apply to every participant and event.

Registration and reporting

- Wear school uniform and carry a valid school ID. Team composition must match registration details.
- Report at 8:15 a.m.; bring your completed robot, model or drone, controllers, batteries and spares. The school will not provide construction parts.

Inspection and equipment

- Report to your event officials for technical inspection before competing. Size, weight, power, control and safety requirements will be checked.
- Weight limits include the battery. A team receives a chance to correct mandatory specification failures before its final inspection.
- Repairs or changes affecting compliance require reinspection.
- Use only the equipment and power sources allowed in your event. Secure batteries, wiring, propellers and moving parts.
- Charge batteries only in the supervised area identified by officials. No brand restriction applies to compliant equipment.

Fair play and safety

- Only registered team members may operate equipment during official attempts. Follow the judge or referee start signal and venue instructions.
- Keep clear of active arenas; retrieve equipment only with permission. Judges may stop play for safety, an arena reset or technical verification.
- Unsafe equipment may return only after correction and clearance.

Penalties and disqualification

- Ordinary mistakes, missed course elements and breakdowns receive the event penalties or an incomplete result. They do not automatically cause disqualification.
- Minor conduct violations receive a warning; event time, point or restart penalties apply where specified.
- Disqualification is limited to mandatory specification non-compliance after a chance to correct; prohibited equipment or power; cheating or interference; unsafe conduct; intentional damage; serious or repeated rule violations; or misconduct. Officials record the reason.

- A safety stop can be immediate.

Official results

- Results use recorded goals, distances, object placements or adjusted times. For timed events, adjusted time equals elapsed time plus penalties; the best completed official run counts.
- Completed runs rank ahead of incomplete runs. Equal official run counts apply within each category.
- Ask the Event Judge promptly about a recorded score or penalty. The judge applies these rules consistently; the final decision is binding.

3. Balloon Car Race

Eligibility: Grades 3-5. Team: 1-3 students. Venue: Natyashala.

Build a balloon-powered car that travels the greatest valid distance.

Equipment limits

Requirement	Limit
Starting size	40 cm long x 30 cm wide
Balloon	One standard round latex balloon; maximum 12 inches
Wheels	Must move primarily on wheels; at least 3 functional wheels
Weight	No maximum weight limit
Attempts	3; best valid distance counts

Build and prepare

- Bring your own balloons and spares.
- **Allowed:** safe everyday or recycled building materials and balloon air propulsion.
- A hand/foot pump may be used only for inflation before an attempt; disconnect it before placing the car on the start line.
- The pump cannot form part of propulsion.
- **Not allowed:** motors, batteries, compressed-air bottles, spring launchers, external pushes or hazardous materials.

Before your attempt

- Bring the completed car for inspection.

How the attempts run

- Each team receives three attempts on a straight 10 m x 1.2 m flat lane.
- All teams complete Attempt 1, then Attempt 2, then Attempt 3.
- Teams have a maximum of two minutes between attempts for retrieval, reinflation and minor repairs; no major redesign is allowed.
- Place the complete car behind the start line and release without pushing.
- An invalid launch scores zero for that attempt.

How distance is measured

- Measure straight down-lane progress from the start line to the foremost part of the car at rest.
- If the car crosses a side boundary, measure to the first crossing point.
- The maximum recorded distance is 10 m.

How the winner is decided

Your best valid distance from the three attempts counts.

Tie break

- Compare the second-best valid distance, then hold one tie-break attempt.
- If still tied, award equal rank.

4. Hydraulic Arm

Eligibility: Grades 6-8. Team: 1-3 students. Venue: Einstein Hall.

Use a manually operated hydraulic arm to transfer objects from Zone A to Zone B.

Equipment limits

Requirement	Limit
Base footprint	Maximum 30 cm x 30 cm
Horizontal reach	Maximum 70 cm from base pivot
Overall height	Maximum 75 cm
Actuation	Manual water hydraulics only
Syringe count and volume	Unrestricted
Task time	Maximum 5 minutes

Build and secure the arm

- **Allowed:** water-filled syringes and tubing, operated by hand, and a safe stable arm structure.
- Syringe count and syringe volume are unrestricted; the number of syringes and degrees of freedom are part of student engineering capability.
- **Not allowed:** motors, batteries, compressors, powered actuators or any working fluid other than water.
- Bring the completed arm for inspection; secure it in the marked base area.
- The base may be weighted or temporarily clamped within this area, provided the table is not damaged.

Reference video

Reference video from the invite: https://youtube.com/shorts/GwhSDpXQtL8

Illustrative reference only. Teams are free to create any compliant design.

Table and task setup

- One task, maximum five minutes.
- The table is 1.8 m x 0.9 m.
- Zones A and B are 300 mm squares with centres 600 mm apart; there is no divider.
- Five identical 30 mm foam cubes weighing 5 g each start in the same arrangement for every team.

Start and object recovery

- The judge/timekeeper starts the timer on "Ready - 3, 2, 1, Start."

- Operate only the hydraulic controls after the start; do not touch task objects by hand.
- A cube remaining on the competition table outside Zones A/B may be retrieved by the arm without hand assistance.
- If a cube falls off the table, an official returns it to Zone A while the timer continues.

Scoring and repairs

- Each cube placed fully inside Zone B and released by the arm earns 20 points, up to 100.
- The highest number of successful placements wins.
- Record the time of each successful placement; stop when all five cubes are transferred or five minutes expire.
- A directly hand-moved cube earns no points and is returned by the official.
- A tube disconnection or water leak may be repaired with permission while the clock continues.

Tie break

- For equal object counts, lower time to the last successful placement wins.
- If no object is placed, teams share rank.
- Other unresolved ties use an identical tie-break task if time permits; otherwise equal rank.

5. Line Follower

Eligibility: Middle and Senior. Team: 1-3 students. Venue: Newton Hall.

An autonomous robot follows a black line on a white surface and completes one full lap as quickly as possible.

Equipment limits

Requirement	Limit
Starting size	20 cm x 20 cm x 20 cm
Weight including battery	Maximum 3 kg
Supply voltage	Maximum 16.8 V; onboard electric power
Control	Autonomous
Line	25 mm wide; black on matte white
Official run limit	3 minutes

Build and control rules

- **Allowed:** onboard sensors and autonomous programming.
- **Not allowed:** remote control, human steering, external guiding lights or objects, track marking or damage.

Tracks and practice

- Pass inspection before running.
- The line is 25 mm wide on a matte white surface; restart points are marked.
- Middle and Senior use separate tracks; exact layouts are not shared before event day.
- A one-hour controlled practice and calibration session on the complete official tracks precedes official runs.
- Run counts and order are announced after entries close, with equal runs within each category.

Starting and completing a run

- A designated team member presses the robot start button only on the judge's "Ready - 3, 2, 1, Start" signal.
- After starting, no human or remote intervention is allowed except the permitted restart procedure.
- Timing ends on completion of one full lap.
- Maximum three minutes per official run; a run not completed within this limit is incomplete.

Line loss and restarts

- If the robot loses the line and cannot recover for 10 continuous seconds, the team may request a restart from the last designated restart point: +5 seconds each, maximum two restarts.
- A third required restart ends the run as incomplete.

Ranking

- Lowest adjusted completed lap time wins.
- Incomplete runs rank by the furthest restart point reached, then lower time to that point.

Tie break

- Fewer restarts in the counting run, then a tie-break run if time permits.
- Unresolved ties receive equal rank.

6. Robo Race

Eligibility: Middle and Senior. Team: 1-3 students. Venue: Newton Hall.

Complete the marked obstacle course in the lowest adjusted time.

Equipment limits

Requirement	Limit
Robot size	Maximum 30 cm x 30 cm x 40 cm
Weight including battery	Maximum 5 kg
Supply	Maximum 12 V DC
Control	Wired manual, wireless manual or fully autonomous
Official run limit	5 minutes

Robot and cable rules

- A control cable must not provide power beyond the permitted supply or pull, lift or physically guide the robot.
- **Not allowed:** dangerous sharp projections, flames, liquids, track damage or systems that interfere with another robot.

Course and inspection

- Pass inspection before running.
- The course includes ramps, a bridge, speed breakers, a seesaw, a fixed rough section and curves.
- Obstacle types may be disclosed; exact layout and module order are revealed on event day.
- Follow numbered checkpoints in order.
- Every team in a category uses the same configuration.

How the run works

- Official run counts and order are announced after entries close; equal runs apply within each category.
- Timing starts on the signal and ends at the finish.
- Maximum five minutes per official run; a run not completed within this limit is incomplete.

- All permitted control modes compete in the same ranking, with no autonomy bonus.

Touches and repairs

- An operator touch or recovery restarts the robot at the previous checkpoint and adds 5 seconds.
- Maximum three touches per run; a fourth ends the run as incomplete.
- Repairs or battery reconnection require permission while the clock continues.

Ranking

- Lowest adjusted completed time wins.
- Incomplete runs rank by the furthest checkpoint reached, then lower time to that checkpoint.

Tie break

- Fewer touches or restarts in the counting run, then a tie-break run if time permits.
- Unresolved ties receive equal rank.

7. Robo Soccer

Eligibility: Middle and Senior. Team: 3 students and 3 robots. Venue: Globe Theatre.

Two teams play 3v3. Score more goals than the opponent in four minutes: two two-minute halves with a one-minute halftime. Change ends at halftime.

Equipment limits

Requirement	Limit for each robot
Size	Maximum 40 cm x 40 cm x 40 cm
Weight including battery	Maximum 7 kg
Supply	Maximum 24 V
Control	Wireless manual only

Team, play and possession

- The same limits apply to Middle and Senior.
- Bring exactly three registered robots; all must pass inspection.
- Start with three compliant robots where possible.
- **Allowed:** pushing, kicking and dribbling the ball, and normal contact while contesting it.

- No goalkeeper/defender is mandatory and there is no fixed limit on defenders in the penalty area; positioning is team strategy subject to normal rules.
- Sustained possession is trapping, clamping, carrying or enclosing the ball so it cannot roll freely for more than three seconds and is prohibited.
- Harmful equipment or interference is prohibited.

Field and kickoff

- The active field is 4.5 m x 3.6 m with perimeter walls and green low-pile carpet.
- Goals have 600 mm clear width and height, and 600 mm depth.
- Officials provide a lightweight plastic ball of target diameter 42 mm.
- A goal counts when the entire ball crosses the goal line between the posts.
- For kickoff, a coin toss winner chooses kickoff or side.
- Place the ball at the centre spot, all robots in their own half and the opposition outside the centre circle.
- One designated robot kicks off on the whistle.
- After a goal, the conceding team gets the next kickoff.
- If the ball is trapped against a wall, between robots or not meaningfully moving for five continuous seconds, the referee calls "Reset", places it at the nearest neutral restart point and resumes on the whistle.

Fouls and their consequences

- Minor fouls include deliberate ramming/pushing away from active play, prolonged pinning, blocking an opponent away from the ball, premature movement during a restart and interference with a restart.
- **First minor foul:** warning and ball reset.
- **Repeated or tactical foul:** opponent restart at the nearest marked restart point.
- Dangerous or repeated misconduct may remove the offending robot for the rest of the match.

Robot failure and re-entry

- Repair or re-entry is allowed only at halftime or a referee stoppage, off the field and with permission.
- After an in-match robot failure or damage, a team may repair/re-enter at a permitted stoppage or voluntarily continue 2v3.
- In-match failure does not automatically cause a walkover.

- No fourth replacement robot.

Control links and interference

- Before each match, test the control links.
- Only playing teams may keep controllers/transmitters on near the arena; waiting teams must keep them off.
- Verified interference may pause the match; resume from the same score and time after resolution.

Results and tie break

- More goals wins.
- A tie goes to Golden Goal for up to two minutes, then three alternating penalties per team and paired sudden-death penalties if needed.
- One attacker faces one stationary goalkeeper; other robots stay off the field.
- Start the ball at the 1 m penalty mark; each attempt lasts up to 10 seconds.
- Alternate the first shooting team after each sudden-death pair.

Fixtures

- **Fixture format:** league/pool, knockout or a combination is determined after registration closes based on entries.
- Final fixtures and reporting order are communicated separately.

8. Drone Quidditch

Eligibility: Middle and Senior. Team: 1-3 students. Venue: Basketball Court.

Fly your own electric quadcopter through numbered suspended hoops in order, then land. Both categories use the same course and rules, with separate results.

Equipment limits

Requirement	Limit
Before take-off size	Fit within 45 cm x 45 cm x 45 cm
Weight including battery	Maximum 1.5 kg
Battery	Maximum 4S LiPo / 14.8 V nominal or equivalent
Control	Manual line of sight; no FPV or autonomous navigation
Propeller guards	Mandatory for every participating drone

Equipment and inspection

- Any brand, commercial or self-built, is allowed; no custom build is required.
- Proprietary manufacturer battery packs are allowed if all other specifications comply.
- Inspect secure propellers, battery mounting, control link, disarming, stable hover and landing.
- Compatible propeller guards are mandatory for all participating drones because the arena will not be fully enclosed by a safety net.

Pilot and course

- One designated pilot remains in the pilot box; one spotter may stand beside them.
- Fly five numbered circular hoops with 1.5 m clear internal diameter, at alternating centre heights of 1.2 m and 1.8 m, then land in a 1 m square zone.
- The course contains no slalom.
- Run counts are announced after entries close; both categories receive the same number.

Clearing hoops and completing the run

- A successful hoop requires the complete drone to pass through the opening of the correct numbered hoop in the required direction.
- After missing a hoop, the pilot may immediately return and re-attempt it before attempting the next numbered hoop.
- Once the next hoop is attempted, the pilot cannot return to an earlier hoop.
- Timing starts on the take-off signal and ends when the drone lands and is disarmed.
- Maximum three minutes per official run; a run not completed within this limit is incomplete.

Penalties

Incident	Time penalty
Missed hoop	+10 seconds
Hoop contact	+5 seconds
Ground contact or crash outside landing zone	+5 seconds per incident
Landing outside marked zone	+5 seconds

- Count each distinct incident once; do not add a second crash penalty for the same ground contact.
- One safe recovery from the last completed hoop is allowed with permission while timing continues.
- Disarm before staff retrieve the drone.
- A second crash or an unsafe drone ends the run as incomplete.
- **Not allowed:** crossing the safety boundary, flying toward spectators, contact with people or other drones, or leaving the pilot box during flight.

Completion and ranking

- Lowest adjusted completed time wins.
- A completed run requires take-off, all numbered hoops attempted in sequence with each cleared or recorded as missed, reaching the landing area, landing and disarming.
- A skipped hoop incurs the +10-second missed-hoop penalty.
- Incomplete runs rank by hoops successfully cleared in sequence, then lower time to the last cleared hoop.
- Ties use fewer penalty incidents, then a tie-break run if safe; otherwise equal rank.

9. Registration and contact

Registration steps

1. Complete payment and submit your registration by 27 October 2026.
2. Keep the payment screenshot ready to upload with the form.
3. Enter the correct school, event, category and team members.
4. Each student may register for one event only; schools may enter unlimited teams.

Item	Official information
Fee	Rs. 500 per participant
Deadline	27 October 2026
Registration form	https://forms.gle/H8o6hrePAacv87Sa6
Payment portal	https://events.strawberryfieldshighschool.com/OnlineRegistration/EventsInSFHS.aspx

Contact Strawberry Fields High School

Email: robosummit@sfhs.in

Student Coordinator: Agam Bir Singh Nandha

WhatsApp: +91 98787 02703 (after 3:00 p.m. only)

Website: https://sfhsrobosummit.in/

Address: Strawberry Fields High School, Sector 26, Chandigarh 160019

Technical Coordination (INNVERSE): +91 99151 14333

Before leaving for the event

- Check your event eligibility and team details.
- Pack the completed equipment, controller, batteries and spares; check every technical limit in your event section.
- Bring your school ID and wear school uniform.
- Reach the school for arrival and reporting at 8:15 a.m. on 30 October 2026.